

```

1  (*
2                                     CS51 Lab 19
3                                     2-dimensional cellular automata
4  *)
5  (*
6                                     SOLUTION
7  *)
8
9  module G = Graphics ;;
10
11  (*****
12  Do not change either of the two module type signatures in this
13  file. Doing so will likely cause your code not to compile
14  against our unit tests.
15  *****)
16
17  (*.....
18  Specifying automata
19
20  A cellular automaton (CA) is a square grid of cells each in a
21  specific state. The grid is destructively updated according to an
22  update rule. The evolving state of the CA can be visualized by
23  displaying the grid in a graphics window.
24
25  In this implementation, a particular kind of automaton, with its
26  state space and update function, is specified by a module satisfying
27  the 'AUT_SPEC' signature. *)
28
29  module type AUT_SPEC =
30    sig
31      (* Automaton parameters *)
32      type state                (* the space of possible cell states *)
33      val initial : state       (* the initial (default) cell state *)
34      val grid_size : int       (* width and height of grid in cells *)
35      val update : (state array array) -> int -> int -> state
36                                (* update grid i j -- returns the new
37                                state for the cell at position 'i, j'
38                                in 'grid' *)
39      val random_state : unit -> state
40                                (* returns a random state *)
41      (* Rendering parameters *)
42      val name : string         (* a display name for the automaton *)
43      val side_size : int       (* width and height of cells in pixels *)
44      val cell_color : state -> G.color

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45                                     (* color for each state *)
46     val legend_color : G.color (* color to render legend *)
47     val font : string option    (* optional font for legend as per
48                                   Graphics module *)
49     val render_frequency : int  (* how frequently grid is rendered
50                                   (in ticks) *)
51 end ;;
52
53 (*.....
54 Automata functionality
55
56 Implementations of cellular automata provides for the following
57 functionality:
58
59 * A current (mutable) grid that can be modified ('replace_grid').
60
61 * Creating additional fresh grids ('fresh_grid').
62
63 * Initializing the graphics window in preparation for rendering
64   ('graphics_init').
65
66 * Mapping the automaton's update function simultaneously over all
67   cells of the current grid ('update_grid').
68
69 * Running the automaton using a particular update function, and
70   rendering it as it evolves ('run_grid').
71
72 as codified in the 'AUTOMATON' signature.
73
74 Note the difference between the argument structure of functions in
75 'life.ml', where the grid is passed in as argument to several
76 functions, as compared to the approach here, where there is a single
77 mutable current grid that is updated by the functions, so that no
78 grid argument is required.
79 *)
80
81 module type AUTOMATON =
82 sig
83   (* state -- Possible states that a grid cell can be in *)
84   type state
85
86   (* grid -- 2D grid of cell states *)
87   type grid = state array array
88
89   (* current_grid -- The current grid of the appropriate size as per
90     the spec, which is initialized with all cells being in the

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91         initial state. It can be modified directly or by `replace_grid`
92         or updated with the automaton's update rule. *)
93     val current_grid : grid
94
95     (* fresh_grid () -- Returns a fresh grid of the appropriate size
96        as per the spec, initialized with all cells being in the
97        initial state. *)
98     val fresh_grid : unit -> grid
99
100    (* random_grid () -- Returns a fresh grid of the appropriate
101       size as per the spec, initialized with cells in random
102       states. *)
103    val random_grid : unit -> grid
104
105    (* replace_grid new_grid -- Destructively changes the current grid
106       ('current_grid', above) to 'new_grid'. *)
107    val replace_grid : grid -> unit
108
109    (* graphics_init () -- Initialize the graphics window to the
110       correct size and other parameters. Auto-synchronizing is off,
111       so our code is responsible for flushing to the screen upon
112       rendering. *)
113    val graphics_init : unit -> unit
114
115    (* update_grid () -- Updates the current grid one "tick" by
116       updating each cell simultaneously as per the CA's update
117       rule. *)
118    val update_grid : unit -> unit
119
120    (* run_grid grid -- Initializes the current grid with `grid` and
121       repeatedly updates it using the automaton's update rule,
122       rendering the grid state to the graphics window until a key is
123       pressed. Assumes graphics has been initialized with
124       `graphics_init`. *)
125    val run_grid : grid -> unit
126    end ;;
127
128    (*.....
129       Implementing automata based on a specification
130
131       Given an automaton specification (satisfying `AUT_SPEC`), the
132       `Automaton` functor delivers a module satisfying `AUTOMATON` that
133       implements the specified automaton. *)
134
135    module Automaton (Spec : AUT_SPEC)
136        : (AUTOMATON with type state = Spec.state) =

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137 struct
138     type state = Spec.state
139     type grid = Spec.state array array
140
141     (* fresh_grid () -- See module type documentation *)
142     let fresh_grid () : unit = grid =
143         Array.make_matrix Spec.grid_size Spec.grid_size Spec.initial
144
145     (* current -- The current grid, which is destructively updated
146        after each tick; initialized with the initial state. *)
147     let current_grid =
148         fresh_grid ()
149
150     let random_grid () : grid =
151         let mat = fresh_grid () in
152         for i = 0 to Spec.grid_size - 1 do
153             for j = 0 to Spec.grid_size - 1 do
154                 mat.(i).(j) <- Spec.random_state ()
155             done
156         done;
157         mat ;;
158
159     (* replace_grid -- See module type documentation *)
160     let replace_grid (new_grid : grid) : unit =
161         for i = 0 to Spec.grid_size - 1 do
162             for j = 0 to Spec.grid_size - 1 do
163                 current_grid.(i).(j) <- new_grid.(i).(j)
164             done
165         done
166
167     (* graphics_init -- See module type documentation *)
168     let graphics_init () : unit =
169         G.open_graph "";
170         G.resize_window (Spec.grid_size * Spec.side_size)
171             (Spec.grid_size * Spec.side_size);
172         (match Spec.font with
173          | None -> ()
174          | Some fontspec -> G.set_font fontspec);
175         G.auto_synchronize false
176
177     (* update_grid -- See module type definition. *)
178     let update_grid =
179         (* use a single static temp grid across invocations *)
180         let temp_grid = fresh_grid () in
181         fun () -> for i = 0 to Spec.grid_size - 1 do
182             for j = 0 to Spec.grid_size - 1 do

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183         temp_grid.(i).(j) <- Spec.update current_grid i j
184     done
185 done;
186 replace_grid temp_grid
187
188 (* render_grid grid legend -- Renders the 'grid' to the already
189    initialized graphics window including the textual 'legend' *)
190 let render_grid (grid : grid) (legend : string) : unit =
191     (* draw the grid of cells *)
192     for i = 0 to Spec.grid_size - 1 do
193         for j = 0 to Spec.grid_size - 1 do
194             G.set_color (Spec.cell_color grid.(i).(j));
195             G.fill_rect (i * Spec.side_size) (j * Spec.side_size)
196                 (Spec.side_size - 1) (Spec.side_size - 1)
197         done
198     done;
199     (* draw the legend *)
200     G.moveto (Spec.side_size * 2) (Spec.side_size * 2);
201     G.set_color Spec.legend_color;
202     G.draw_string legend;
203     (* flush to the screen *)
204     G.synchronize ()
205
206 (* run_grid -- See module type documentation *)
207 let run_grid (initial_grid : grid) : unit =
208     replace_grid initial_grid;
209     let tick = ref 0 in
210     render_grid current_grid (Printf.sprintf "%s: tick %d" Spec.name !tick);
211     ignore (G.read_key ()); (* pause at start until key is pressed *)
212     while not (G.key_pressed ()) do
213         update_grid ();
214         tick := succ !tick;
215         if !tick mod Spec.render_frequency = 0 then
216             render_grid current_grid (Printf.sprintf "%s: tick %d" Spec.name !tick);
217         done;
218         ignore (G.read_key ()); (* clear the keypress *)
219         ignore (G.read_key ()); (* await another keypress *)
220         G.close_graph ();
221 end ;;

```